Chapter 4

Data Link Layer

Chapter 4 Outline

4.1 - Introduction

4.2 - Media Access Control

- Contention, Controlled Access, Relative Performance

4.3 - Error Control

- Sources of Errors, Error Prevention, Error Detection, Error Correction via Retransmission, Forward Error Correction

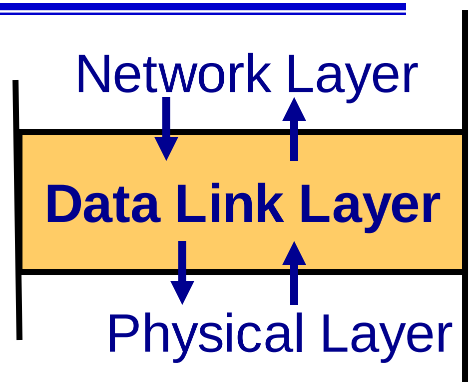
4.4 - Data Link Protocols

- Asynchronous Transmission, Synchronous Transmission

4.5 - Transmission Efficiency

4.6 – Implications for Management

4.1 Introduction



- Responsible for moving messages from one device to another
- Controls the way messages are sent on media
- Organizes physical layer bit streams into coherent messages for the network layer
- Major functions of a data link layer protocol
  - Media Access Control
    - Controlling when computers transmit
  - Error Control
    - Detecting and correcting transmission errors
  - Message Delineation
    - Identifying the beginning and end of a message

4.2 Media Access Control (MAC)

- Controlling when and what computer transmit
  - Important when more than one computer wants to send data at the same time over the same, shared circuit
    - Point-to-point half duplex links
      - computers take turns
    - Multipoint configurations
      - Ensure that no two computers attempt to transmit data at the same time
- Two possible approaches
  - Contention based access
  - Controlled access

Controlled Access

- Controlling access to shared resources
  - Acts like a stop light
- Commonly used by mainframes (or its front end processor)
  - Determines which circuits have access to mainframe at a given time
- Also used by some LAN protocols
  - Token ring, FDDI
- Major controlled access methods
  - Access request and polling

Access Request

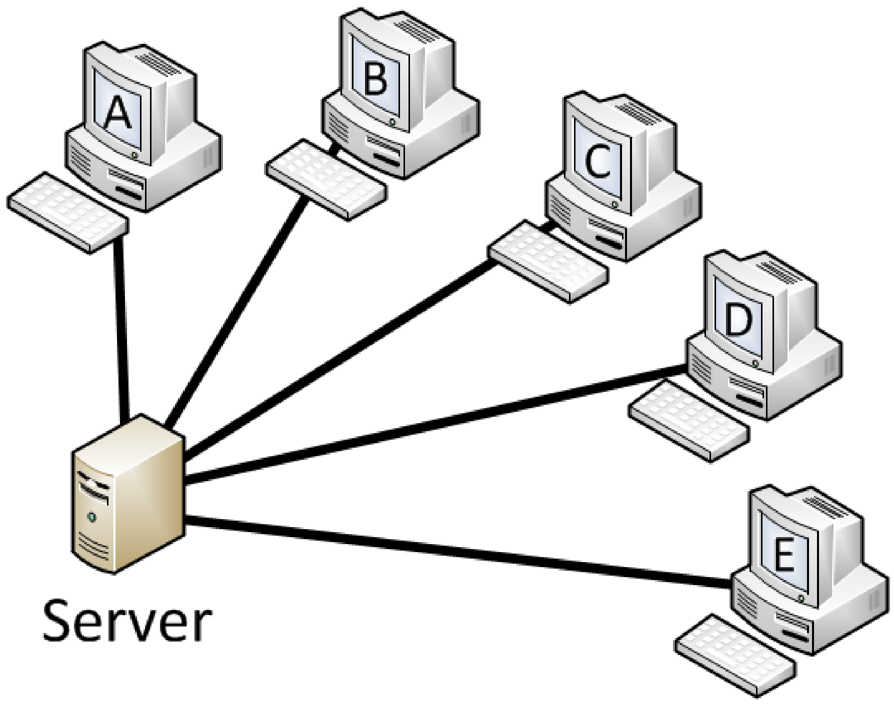
- Clients wanting to transmit data first send a request to the device controlling the circuit
- The central device will grant permission for one device at a time to transmit

Polling

- Process of transmitting to a client only if asked and/or permitted
  - Client stores the information to be transmitted
  - Server (periodically) polls the client if it has data to send
  - Client, if it has any, sends the data
  - If no data to send, client responds negatively, and server asks the next client
- Types of polling
  - Roll call polling
  - Hub polling (also called token passing)

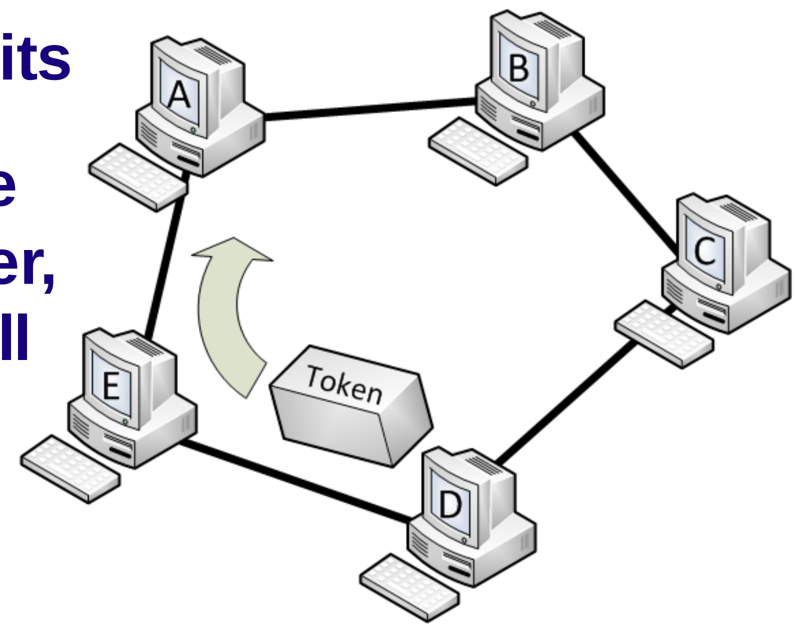
Roll Call Polling

- Check each client (consecutively and periodically) to see if it wants to transmit : A, B, C, D, E, A, B, …
- Clients can also be prioritized so that they are polled more frequently: A, B, A, C, A, D, A, E, A, B, ..
- Involves waiting: Poll and wait for a response
- Needs a timer to prevent lock-up (by client not answering)



Hub Polling (Token Passing)

- One computer starts the poll:
  - sends message (if any) then
  - passes the token to the next computer
  - token is a unique series of bits
- Continues in sequence until the token reaches the first computer, which starts the polling cycle all over again

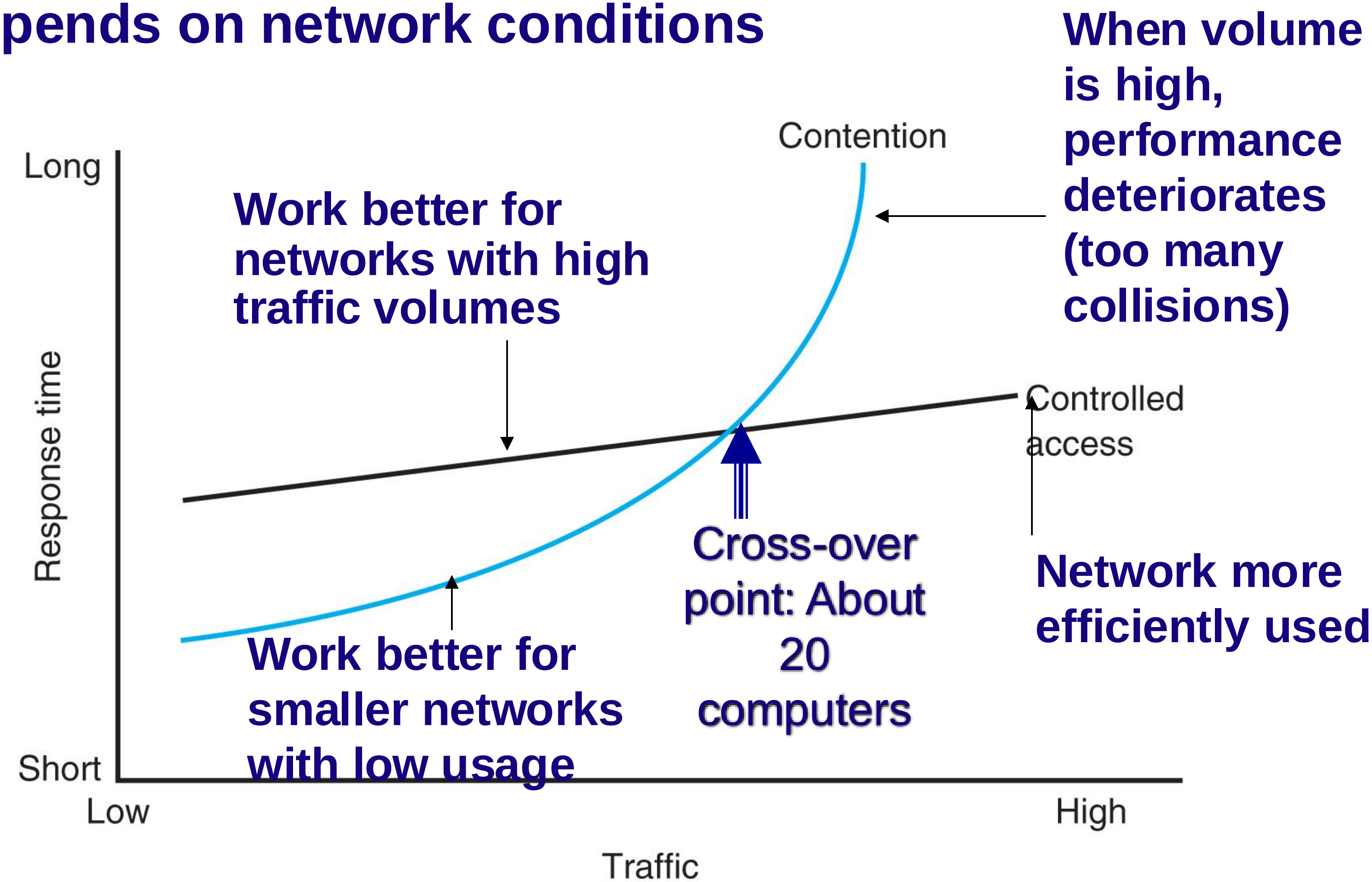


Contention

- Transmit whenever the circuit is free
- Collisions
  - Occur when more than one computer transmits at the same time
  - Need to determine which computer is allowed to transmit first after the collision
- Used commonly in Ethernet LANs
- Can be problematic in heavy usage networks

Relative Performance

Depends on network conditions



4.3 - Error Control

- Handling of network errors caused by problems in transmission
  - Network errors
    - Can be a bit value change during transmission
    - Controlled by network hardware and software
  - Human errors:
    - Can be a mistake in typing a number
    - Controlled by application programs
- Categories of Network Errors
  - Corrupted (data that has been changed)
  - Lost data (cannot find the data at all)

Error Control (Cont.)

- **Error Rate**
  - **1 bit error in n bits transmitted, e.g., 1 in 500,000**
- **Burst error** (more common)
  - **Many bits are corrupted at the same time**
  - **Errors not uniformly distributed**
    - **e.g., 100 in 50,000,000 → 1 in 500,000**

Sources of Errors

- Line noise and distortion – major cause
  - More likely on electrical media
  - Undesirable electrical signal
  - Introduced by equipment and natural disturbances
  - Degrades performance of a circuit
  - Manifestation
    - Extra bits
    - “Flipped” bits
    - Missing bits

Major Functions of Error Control

- Error prevention
- Error detection
- Error correction

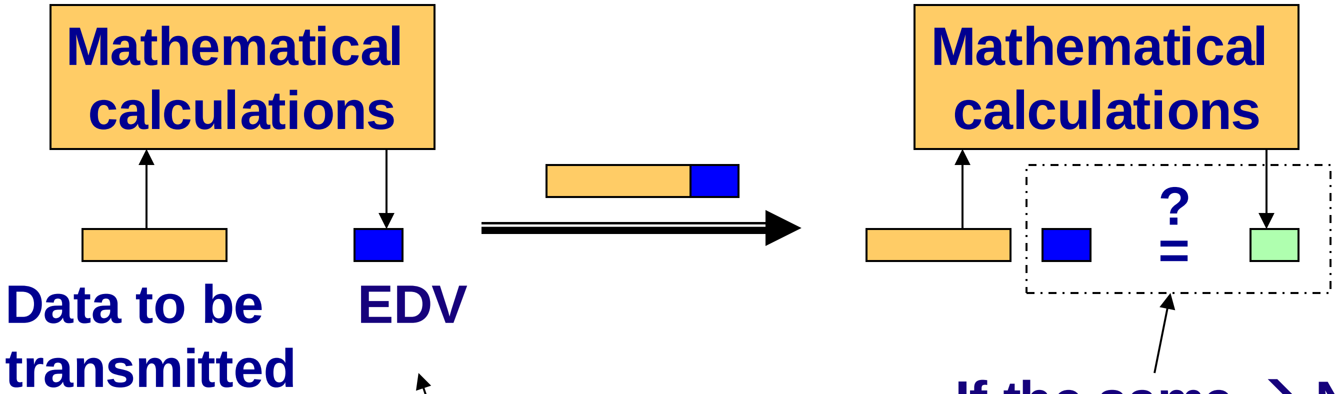
Sources of Errors and Prevention

Source of error	Cause	Prevention
White noise	Movement of electrons (thermal energy)	Increase signal strength (increase SNR)
Impulse noise	Sudden increases in electricity (e.g., lightning, power surges)	Shield or move the wires
Cross-talk	Multiplexer guard bands are too small or wires too close together	Increase the guard bands, or move or shield the wires
Echo	Poor connections (causing signal to be reflected back to the source)	Fix the connections, or tune equipment
Attenuation	Gradual decrease in signal over distance (weakening of a signal)	Use repeaters or amplifiers
Intermodulation noise	Signals from several circuits combine	Move or shield the wires

Error Detection

Sender calculates an Error Detection Value (EDV) and transmits it along with data

Receiver recalculates EDV and checks it against the received EDV



Larger the size, better error detection (but lower efficiency)

- If the same → No errors in transmission
- If different → Error(s) in transmission

Error Detection Techniques

- Parity checks
- Checksum
- Cyclic Redundancy Check (CRC)

Parity Checking

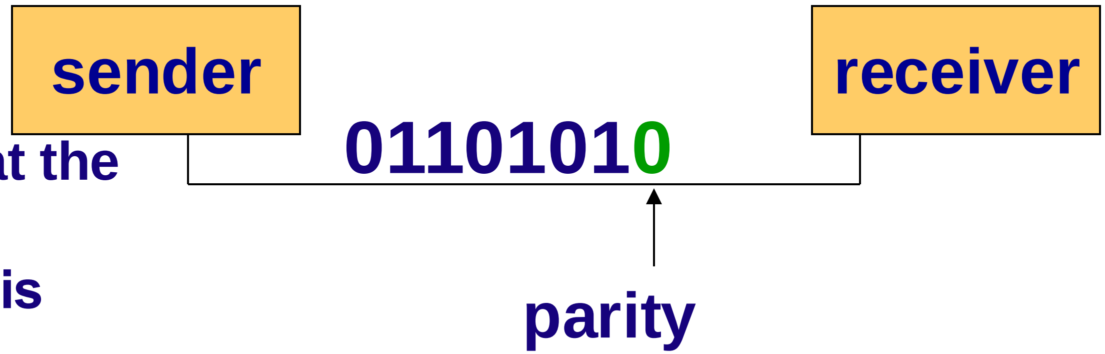
- One of the oldest and simplest
- A single bit added to each character
  - Even parity: number of 1’s remains even
  - Odd parity: number of 1’s remains odd
- Receiving end recalculates parity bit
  - If one bit has been transmitted in error the received parity bit will differ from the recalculated one
- Simple, but doesn’t catch all errors
  - If two (or an even number of) bits have been transmitted in error at the same time, the parity check appears to be correct
  - Detects about 50% of errors

Examples of Using Parity

To be sent: Letter V in 7-bit ASCII: 0110101

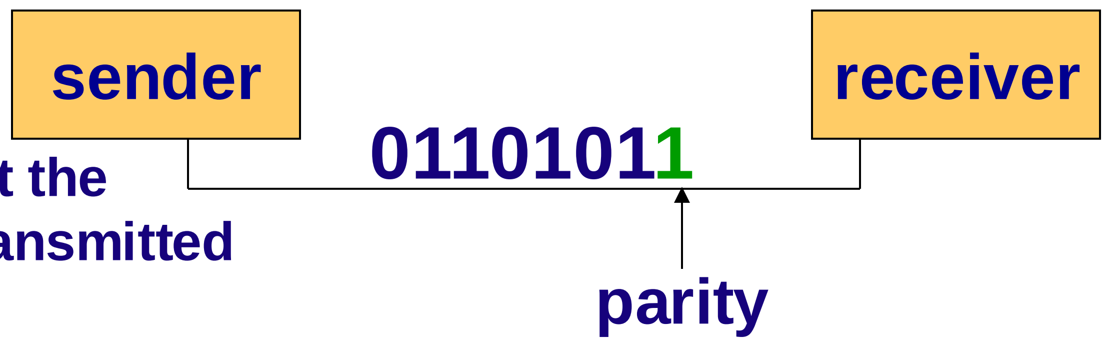
EVEN parity

Add a bit so that the number of all transmitted 1's is EVEN



ODD parity

Add a bit so that the number of all transmitted 1's is ODD



Checksum

- A checksum (usually 1 byte) is added to the end of the message
- It is 95% effective
- Method:
  - Add decimal values of each character in the message
  - Divide the sum by 255
  - The remainder is the checksum value

Cyclic Redundancy Check (CRC)

$$P / G = Q + R / G$$

Example:
P = 58
G = 8
Q = 7
R = 2

Message (treated as one long binary number)

A fixed number (determines the length of the R)

Quotient (whole number)

Remainder:
- added to the message as EDV
- could be 8 bits, 16 bits, 24 bits, or 32 bits long
- CRC16 has R of 16 bits

- Most powerful and most common
- Detects 100% of errors (if number of errors <= size of R)
  - Otherwise: CRC-16 (99.998%) and CRC-32 (99.9999%)

Error Correction

- **Once detected, the error must be corrected**
- **Error correction techniques**
  - **Retransmission (or, backward error correction)**
    - **Simplest, most effective, least expensive, most commonly used**
    - **Corrected by retransmission of the data**
      - **Receiver, when detecting an error, asks the sender to retransmit the message**
    - **Often called Automatic Repeat reQuest (ARQ)**
  - **Forward Error Correction**
    - **Receiving device can correct incoming messages without retransmission**

Automatic Repeat reQuest (ARQ)

- Process of requesting a data transmission be resent
- Main ARQ protocols
  - Stop and Wait ARQ (A half duplex technique)
    - Sender sends a message and waits for acknowledgment, then sends the next message
    - Receiver receives the message and sends an acknowledgement, then waits for the next message
  - Continuous ARQ (A full duplex technique)
    - Sender continues sending packets without waiting for the receiver to acknowledge
    - Receiver continues receiving messages without acknowledging them right away

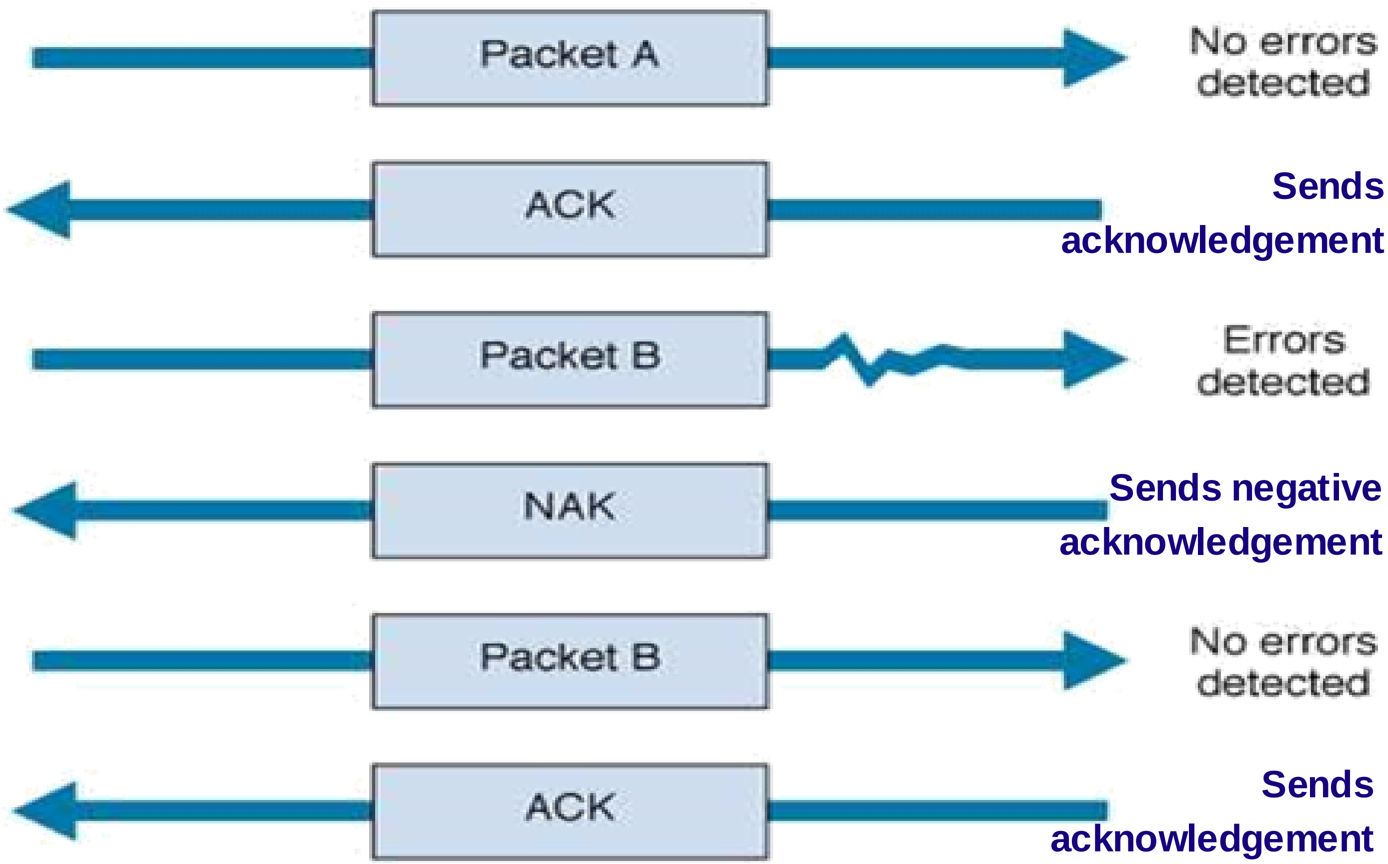
Stop and Wait ARQ

Sender | **Receiver**

Sends Packet A, then waits to hear from receiver.

Sends the next packet (B)

Resends the packet again

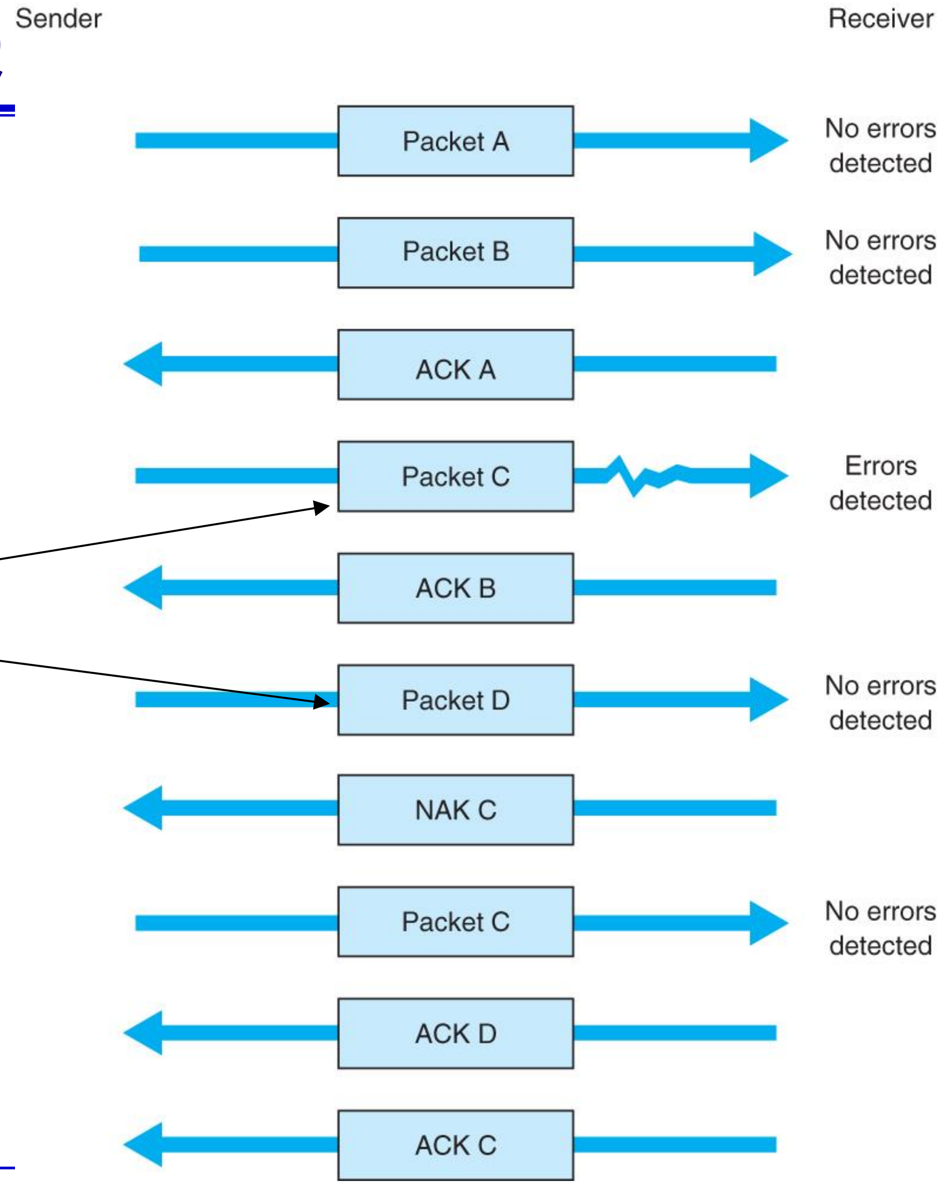


Continuous ARQ

Sender sends packets continuously without waiting for receiver to acknowledge

Notice that acknowledgments now identify the packet being acknowledged.

Receiver sends back a NAK for a specific packet to be resent.

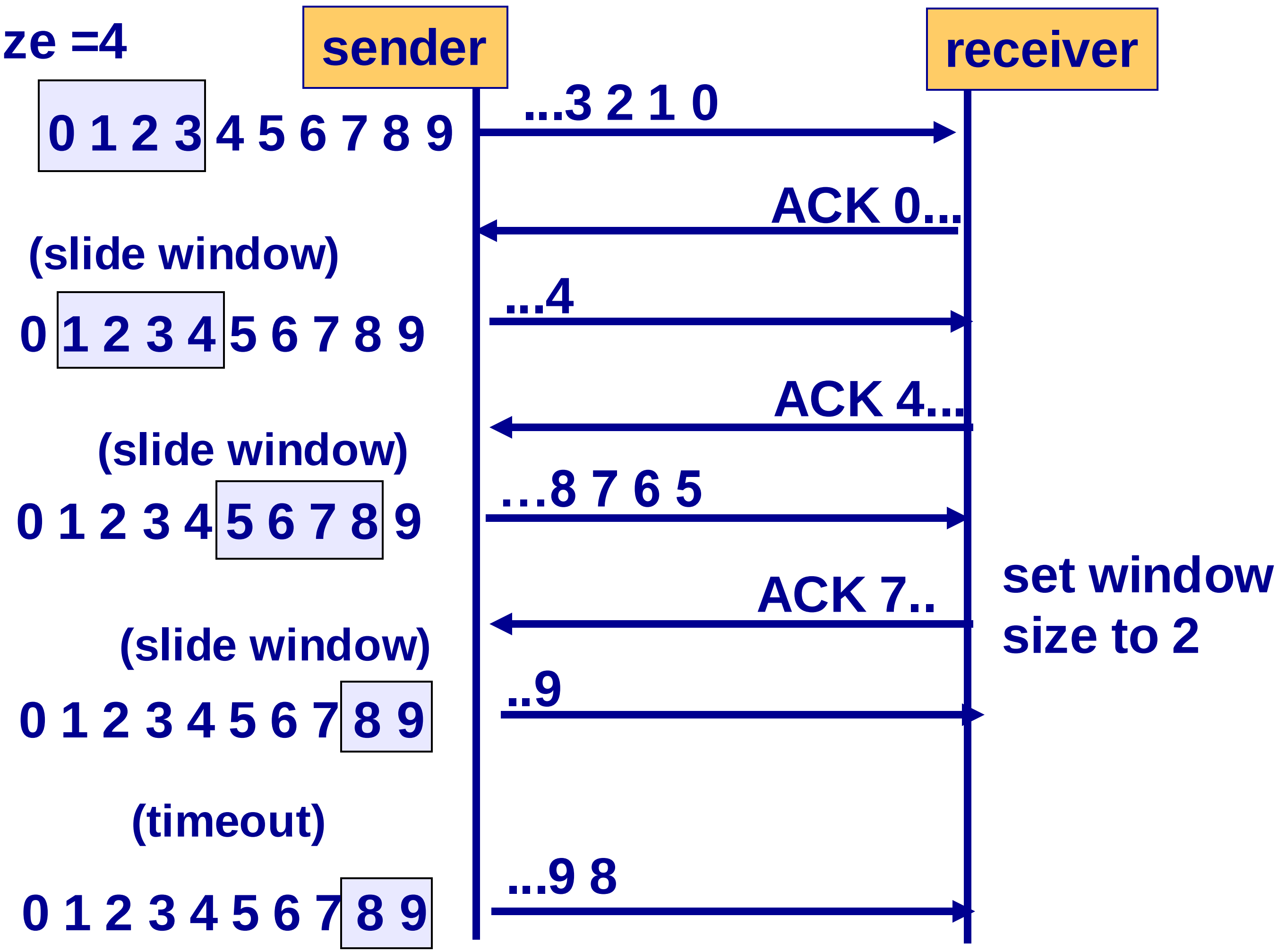


Flow Control with ARQ

- Ensuring that sender is not transmitting too quickly for the receiver
  - Stop-and-wait ARQ
    - Receiver sends an ACK or NAK when it is ready to receive more packets
  - Continuous ARQ:
    - Both sides agree on the size of the "sliding window"
      - Number of messages that can be handled by the receiver without causing significant delays

Flow Control Example

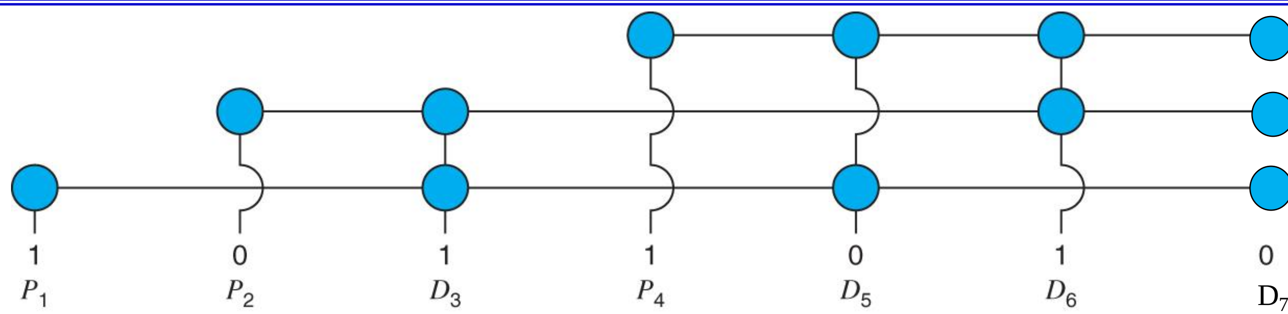
window size =4



Forward Error Correction (FEC)

- Receiving device can correct incoming messages itself (without retransmission)
- Requires extra corrective information
  - Sent along with the data
  - Allows data to be checked and corrected by the receiver
  - Amount of extra information: usually 50-100% of the data
- Used in the following situations:
  - One way transmissions (retransmission not possible)
  - Transmission times are very long (satellite)
  - In this situation, relatively insignificant cost of FEC

Hamming Code – An FEC Example



Checking relations between parity bits (P) and data bits (D)

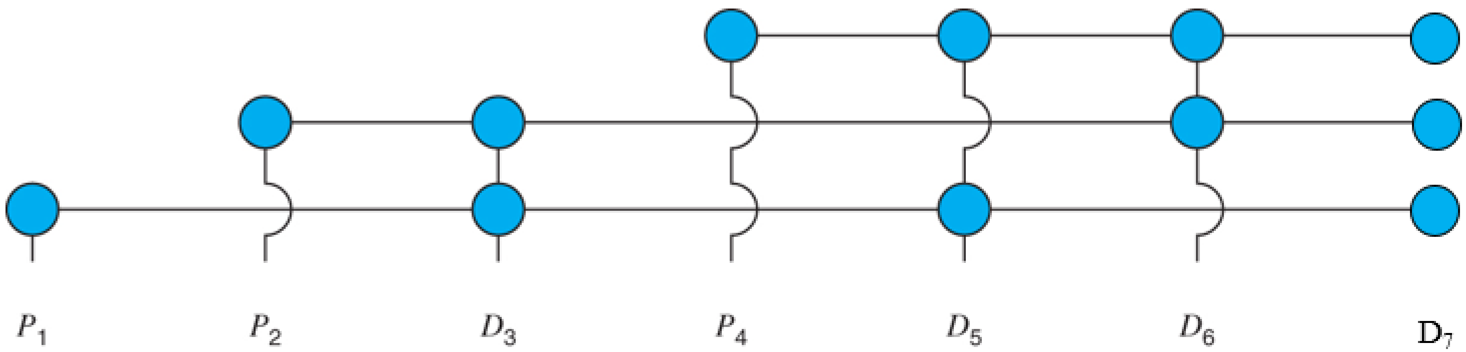
Each data bit figures into three EVEN parity bit calculations

If any one bit (parity or data) changes → change in data bit can be detected and corrected

√ = Corresponding parity check is correct X = Corresponding parity check fails				Determines in which bit the error occured
P_4	P_2	P_1		
√	√	√	→	no error
√	√	X	→	P_1
√	X	√	→	P_2
√	X	X	→	D_3
X	√	√	→	P_4
X	√	X	→	D_5
X	X	√	→	D_6
X	X	X	→	D_7

Interpreting parity bit patterns

Only works for one bit errors



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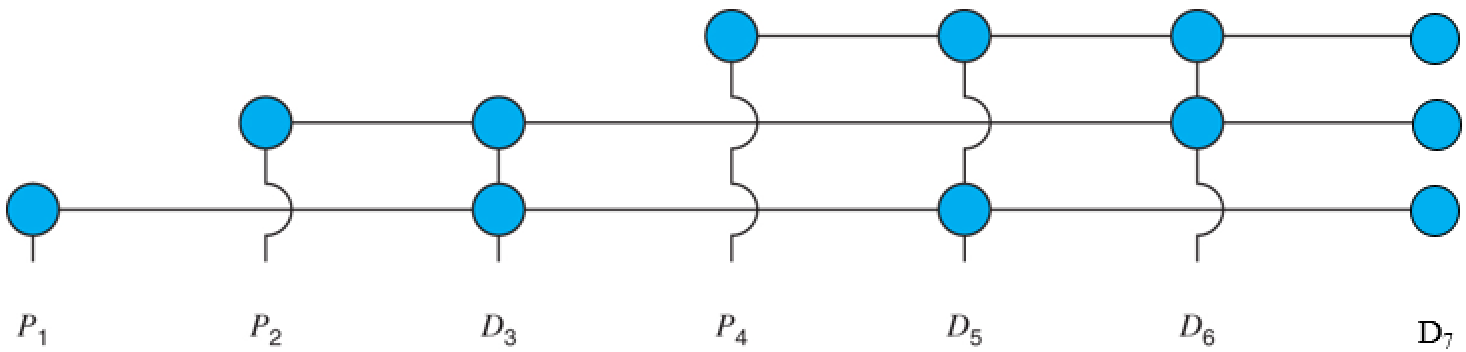
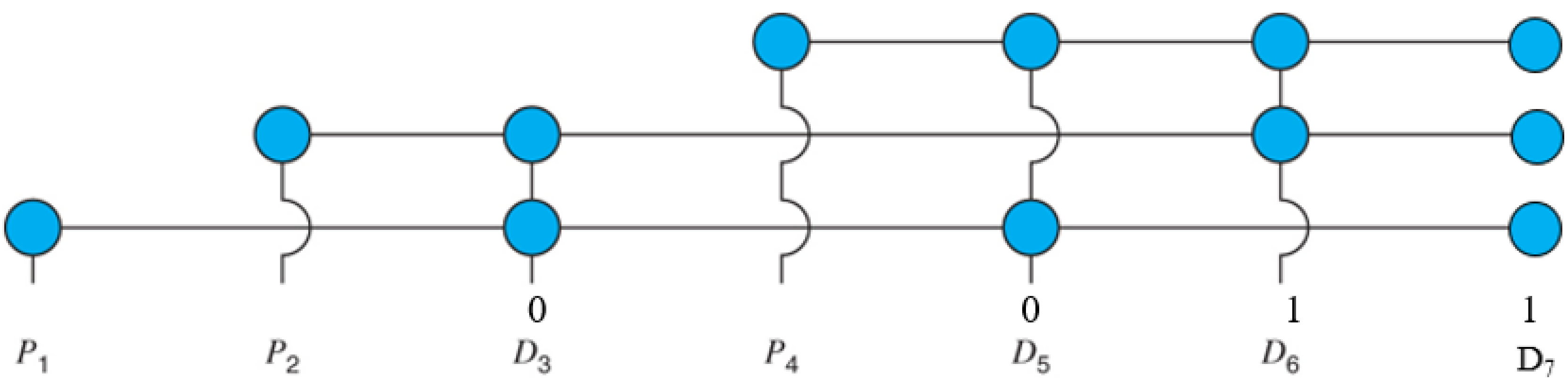
X = Corresponding parity check fails

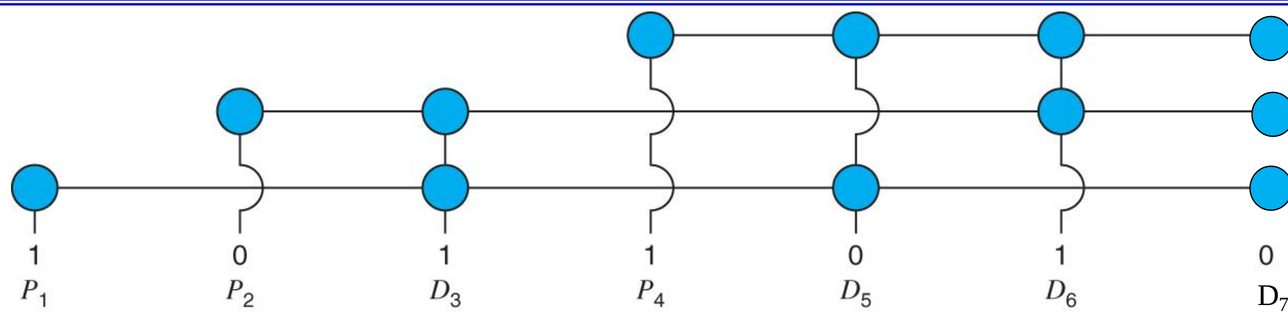
Determines in which bit the error occured

P_4	P_2	P_1		
√	√	√	→	no error
√	√	X	→	P_1
√	X	√	→	P_2
√	X	X	→	D_3
X	√	√	→	P_4
X	√	X	→	D_5
X	X	√	→	D_6
X	X	X	→	D_7


P1
P2
D3
P4
D5
D6
D7


0
0
1
1
P1
P2
D3
P4
D5
D6
D7

Hamming Code – An FEC Example



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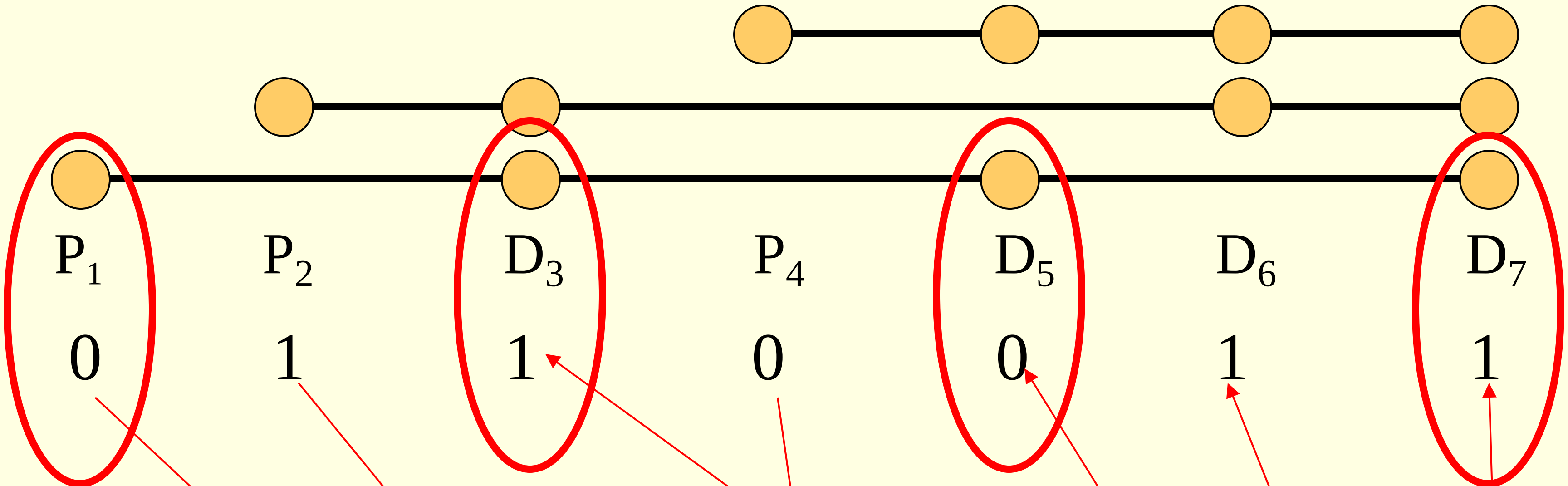
P_4	P_2	P_1	Determines in which bit the error occured
√	√	√	no error
√	√	X	P_1
√	X	√	P_2
√	X	X	D_3
X	√	√	P_4
X	√	X	D_5
X	X	√	D_6
X	X	X	D_7

Interpreting parity bit patterns

Only works for one bit errors

Hamming Example

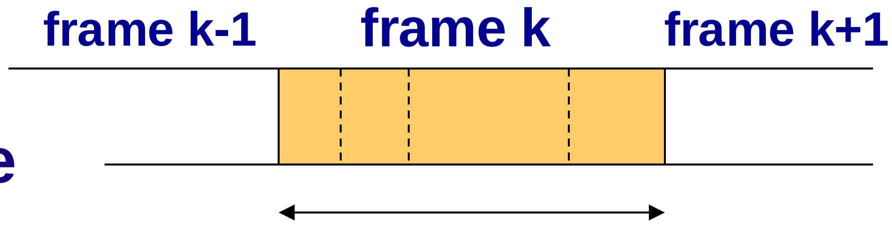


Assuming even parity and given the data bits 1, 0, 1, 1

What are the parity bits? 0, 1, 0

4.4 Data Link Protocols

- Classification
  - Asynchronous transmission
  - Synchronous transmission
- Differ by
  - Message delineation
  - Frame length
  - Frame field structure

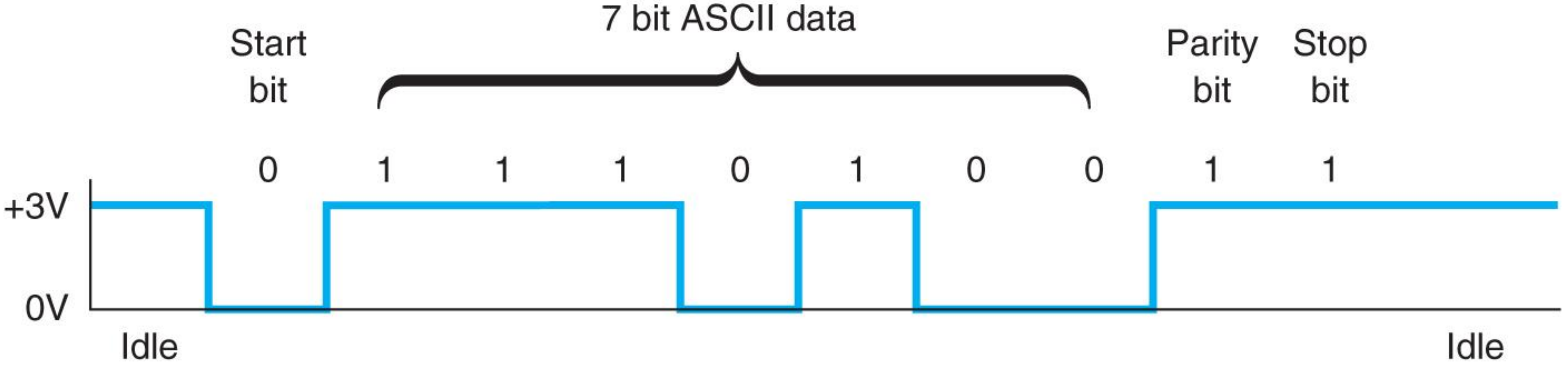


Asynchronous Transmission

Start bit used by the receiver for separating characters and for synch.

Each character is sent independently

Stop bits sent between transmissions (a series of stop bits)



Asynchronous File Transfer

- Used on
  - Point-to-point asynchronous circuits
  - Typically over phone lines via modem
  - Computer to computer for transfer of data files
- Sometimes called Start/Stop Transmission
- Characteristics of file transfer protocols
  - Designed to transmit error-free data
  - Group data into blocks to be transmitted (rather sending character by character)
- Popular File transfer Protocols
  - Xmodem, Zmodem, and Kermit

File Transfer Protocols

Xmodem:
- One of the oldest async file transfer protocol
- Uses stop-and-wait ARQ.

Xmodem-CRC:
- uses 1 byte CRC (instead of checksum)

Xmodem-1K:
- Xmodem-CRC + 1024 byte long message field

Zmodem:
- Uses CRC-32 with continuous ARQ
- Dynamic adjustment of packet size

Synchronous Transmission

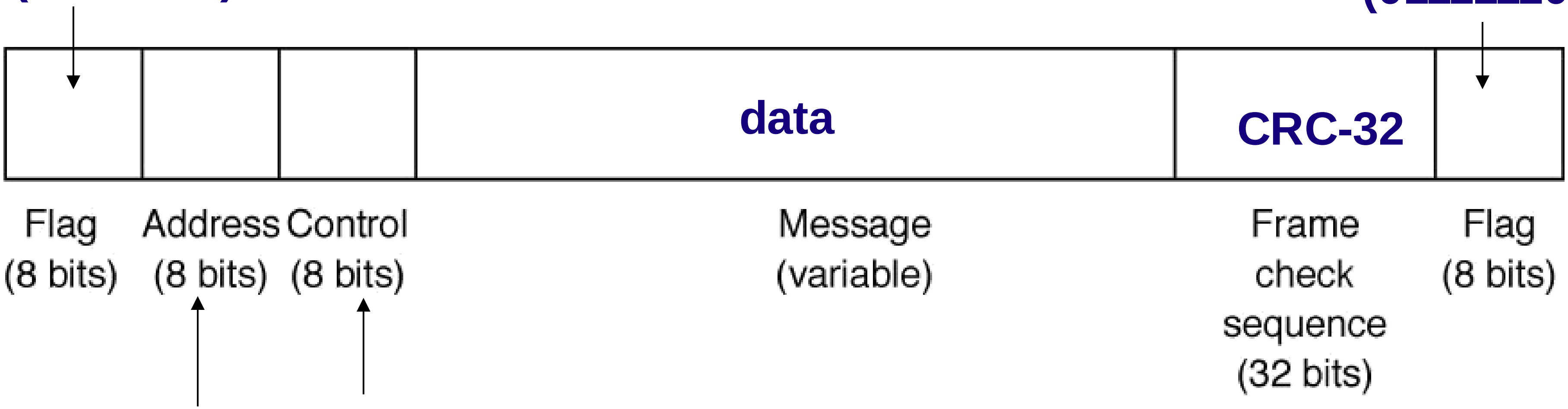
- Data sent in a large block
  - Called a frame or packet
  - Typically about a thousand characters (bytes) long
- Includes addressing information
  - Especially useful in multipoint circuits
- Includes a series of synchronization (SYN) characters
  - Used to help the receiver recognize incoming data
- Synchronous transmission protocols categories
  - Bit-oriented protocols: SDLC, HDLC
  - Byte-count protocols: Ethernet
  - Byte-oriented protocols: PPP

SDLC – Synchronous Data Link Control

- Bit-oriented protocol developed by IBM
- Uses a controlled media access protocol

Beginning (01111110)

Ending (01111110)



Destination Address (8 or 16 bits)

Identifies frame type;

- Information (for transferring of user data)
- Supervisory (for error and flow control)

Transparency Problem of SDLC

- **Problem: Transparency**
  - User data may contain the same bit pattern as the flags (01111110)
  - Receiver may interpret it as the end of the frame and ignores the rest
- **Solution: Bit stuffing (aka, zero insertion)**
  - Sender inserts 0 anytime it detects 11111 (five 1’s)
  - If receiver sees five 1's, checks next bit(s)
    - if 0, remove it (stuffed bit)
    - if 10, end of frame marker (01111110)
    - if 11, error (7 1's cannot be in data)
  - Works but increases complexity

HDLC – High-Level Data Link Control

- Formal standard developed by ISO
- Same as SDLC, except
  - Longer address and control fields
  - Larger sliding window size
  - And more
- Basis for many other Data Link Layer protocols
  - LAP-B (Link Access Protocol – Balanced)
    - Used by X.25 technology
  - LAP-D (Link Access Protocol – Balanced)
    - Used by ISDN technology
  - LAP- F (Used by Frame Relay technology)

Ethernet (IEEE 802.3)

- Most widely used LAN protocol, developed jointly by Digital, Intel, and Xerox, now an IEEE standard
- Uses contention based media access control
- Byte-count data link layer protocol
- No transparency problem
  - uses a field containing the number of bytes (not flags) to delineate frames
- Error correction: optional

Ethernet 802.3ac frame layout

- Used by Virtual LANs; if no vLAN, the field is omitted
- If used, first 2 bytes set to 24,832 (8100H)

- Number of bytes in the message field

- Used to hold sequence number, ACK/NAK, (1 or 2 bytes)

- CRC-32 error detection

Preamble	Start of Frame	Destination Address	Source Address	VLAN Tag	Length	DSAP	SSAP	Control	Data	Frame Check Sequence
7 bytes	1 byte	6 bytes	6 bytes	4 bytes	2 bytes	1 byte	1 byte	1-2 bytes	46-1,500 bytes	4 bytes

repeating pattern of ones and zeros (10101010)

Destination/Source Service Access Point

- conceptual location at which one OSI layer can request the services of another OSI layer
- usually is the network layer protocol

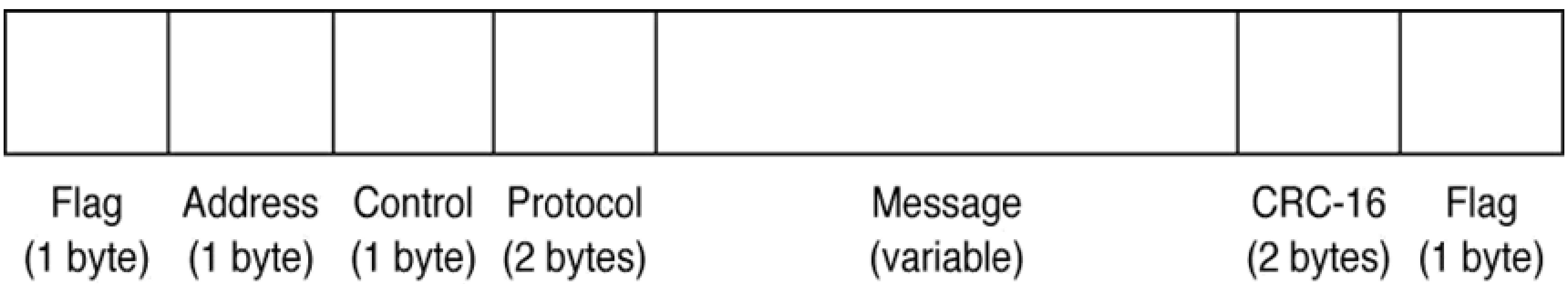
Ethernet II frame layout

specify an ACK frame or the type of network layer packet the frame contains (e.g., IP).

Preamble	Start of Frame	Destination Address	Source Address	Type	Data	Frame Check Sequence
7 bytes	1 byte	6 bytes	6 bytes	2 bytes	46-1,500 bytes	4 bytes

Point-to-Point Protocol (PPP)

- Byte-oriented protocol developed in early 90s
- Commonly used on dial-up lines from home PCs
- Designed mainly for point-to-point phone line (can be used for multipoint lines as well)



(up to 1500 bytes)

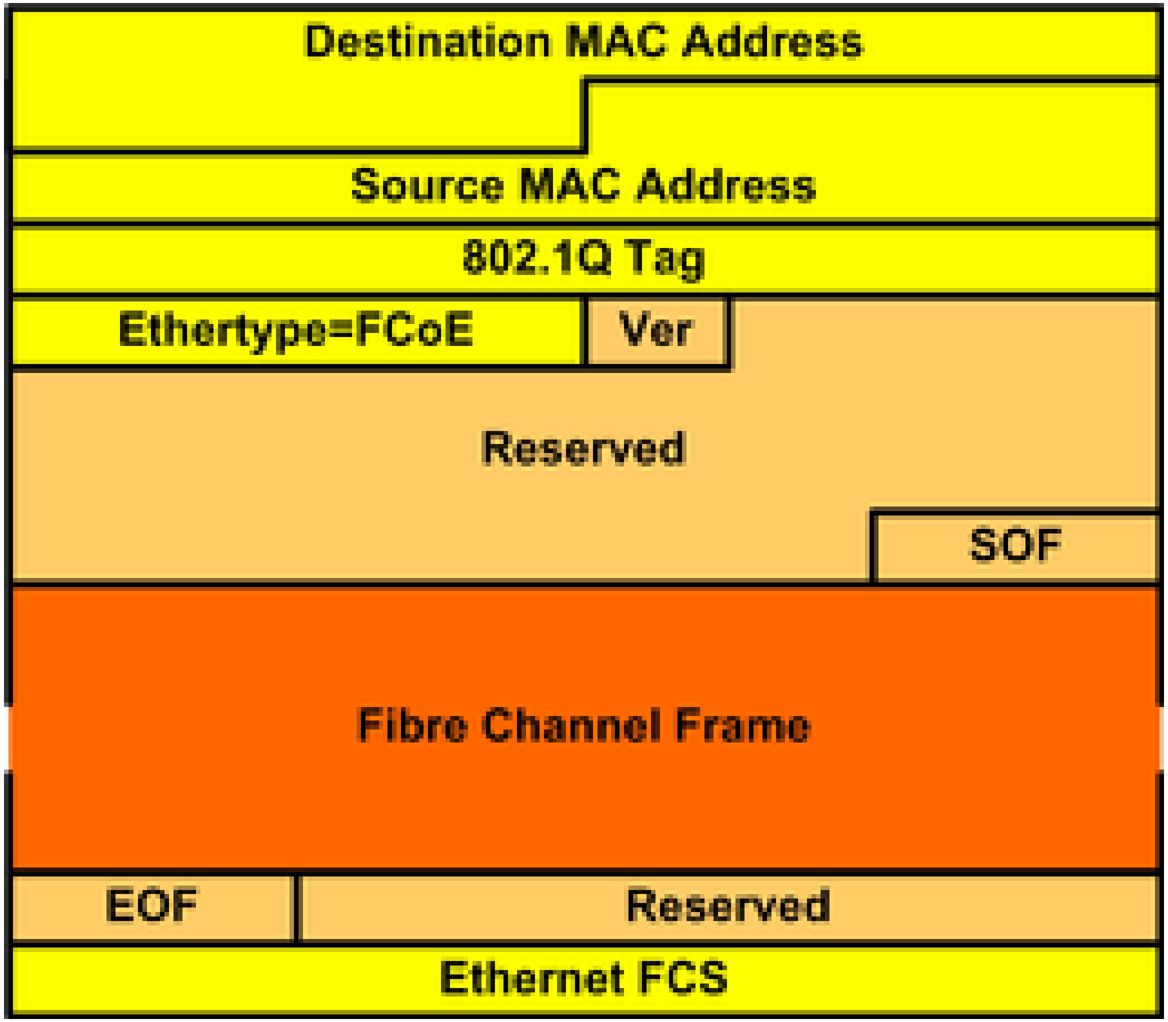
Specifies the network layer protocol used (e.g, IP, IPX)

Data Link Protocol Summary

Protocol	Size	Error Detection	Retransmission	Media Access
Asynchronous Xmission	1	Parity	Continuous ARQ	Full Duplex
File Transfer Protocols				
XMODEM	132	8-bit Checksum	Stop-and-wait ARQ	Controlled Access
XMODEM-CRC	132	8-bit CRC	Stop-and-wait ARQ	Controlled Access
XMODEM-1K	1028	8-bit CRC	Stop-and-wait ARQ	Controlled Access
ZMODEM	*	32-bit CRC	Continuous ARQ	Controlled Access
KERMIT	*	24-bit CRC	Continuous ARQ	Controlled Access
Synchronous Protocols				
SDLC	*	16-bit CRC	Continuous ARQ	Controlled Access
HDLC	*	16-bit CRC	Continuous ARQ	Controlled Access
Token Ring	*	32-bit CRC	Stop-and wait ARQ	Controlled Access
Ethernet	*	32-bit CRC	Stop-and wait ARQ	Contention
SLIP	*	None	None	Full Duplex
PPP	*	16-bit CRC	Continuous ARQ	Full Duplex

*** Varies depending on message length.**

Fibre Channel over Ethernet, FCoE



4.5 Transmission Efficiency

- An objective of the network:
  - Move as many bits as possible with minimum errors
    → higher efficiency and lower cost
- Factors affecting network efficiency:
  - Characteristics of circuit (error rate, speed)
  - Speed of equipment, Error control techniques
  - Protocol used
    - Information bits (carrying user information)
    - Overhead bits (used for error checking, frame delimiting, etc.)

$$= \frac{\text{Total number of info bits to be transmitted}}{\text{Total number of bits transmitted}}$$

Transmission Efficiency of Protocols

Async Transmission:

7-bit ASCII (info bits), 1 parity bit, 1 stop bit, 1 start bit

<u>Transmission Efficiency</u> = 7 / 10 → 70%

e.g., V.92 modem with 56 Kbps → 39.2 Kbps effective rate

SDLC Transmission

Assume 100 info characters (800 bits), 2 flags (16 bits)
Address (8 bits), Control (8 bits), CRC (32 bits)

<u>Transmission Efficiency</u> = 800 / 864 → 92.6%

e.g., V.92 modem with 56 Kbps → 51.9 Kbps effective rate

Bigger the message length, better the efficiency

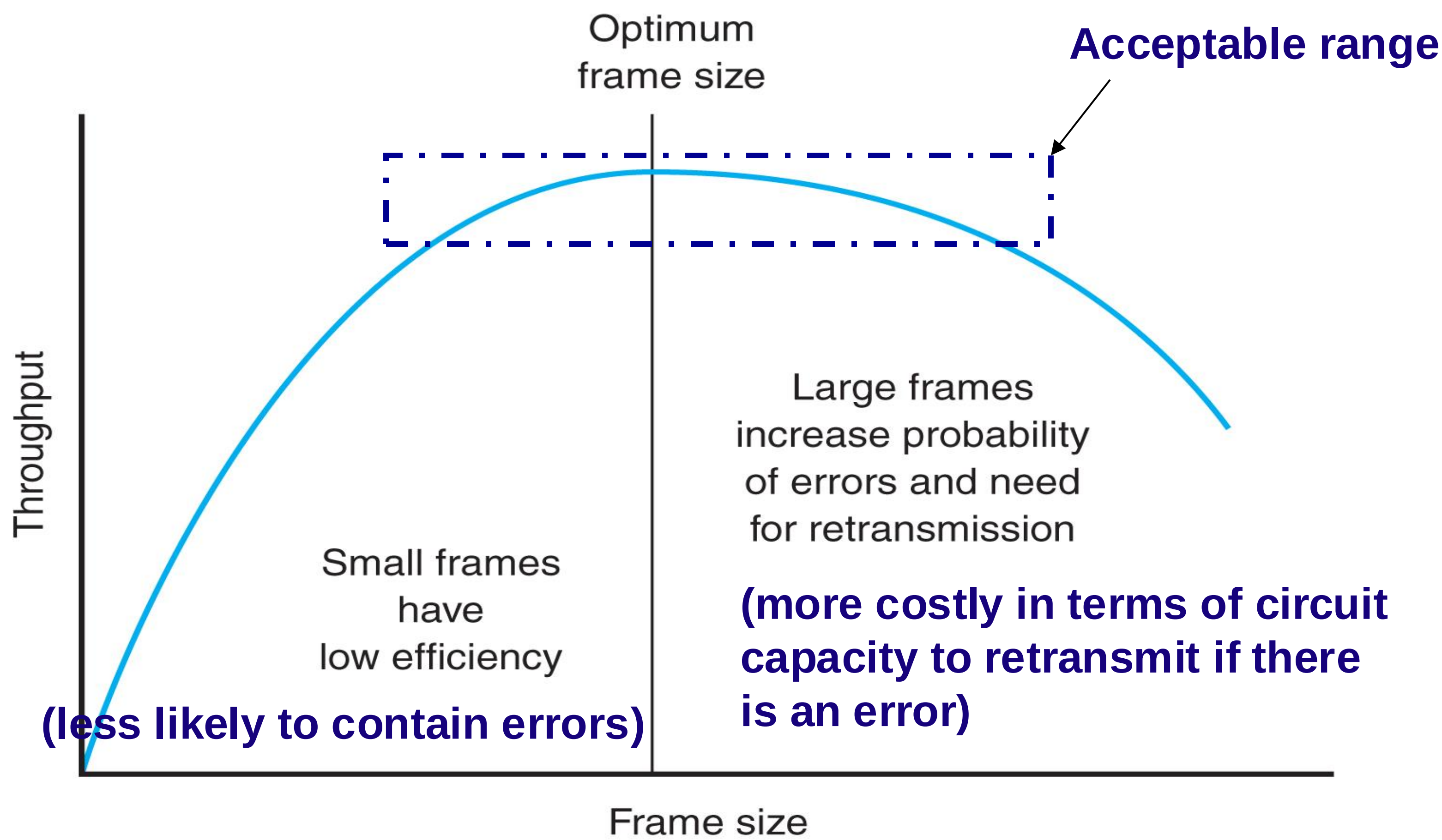
However, large packets likely to have more errors and are more likely to require retransmission → wasted capacity

Throughput

- A more accurate definition of efficiency
- Total number of information bits received per second; takes into account:
  - Overhead bits (as in transmission efficiency)
  - Need to retransmit packets containing errors
- Complex to calculate; depends on:
  - Transmission efficency
  - Error rate
  - Number of retransmission
- Transmission Rate of Information Bits (TRIB)
  - Used as a measurement of throughput

Optimum Packet Size

Trade-off between packet size and throughput



TRIB

FORMULA FOR CALCULATING TRIB

$$\text{TRIB} = \frac{\text{Number of information bits accepted}}{\text{Total time required to get the bits accepted}}$$

$$\text{TRIB} = \frac{K(M - C)(1 - P)}{(M/R) + T}$$

where K = information bits per character

M = frame length in characters

R = data transmission rate in characters per second

C = average number of noninformation characters per frame (control characters)

P = probability that a frame will require retransmission because of error

T = time between frames in seconds, such as modem delay/turnaround time on half duplex and propogation delay on satellite transmission. This is the time required to reverse the direction of transmission from send to receive or receive to send on a half-duplex circuit. It can be obtained from the modem specification book and may be referred to as *reclocking time*.

4.6 Implications for Cyber Security

- MAC address a.k.a physical address for security
  - MAC address filtering to allow certain computers to connect to Wi-Fi network or switch in corporate network.
  - However, MAC address filtering can offer a false sense of security because of MAC address spoofing.
  - MAC address spoofing is a software-enable technique that can change the hardcoded MAC address from factory to any address, and overcome MAC address filtering.

References

- Business Data Communications and Networking 14th Edition
  Jerry Fitzgerald, Alan Dennis, Alexandra Durcikova; John Wiley & Sons, Inc